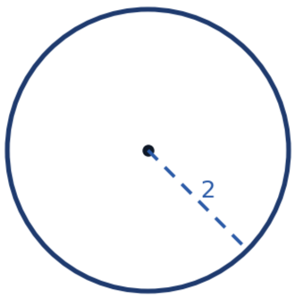
# Polar Curves and Derivatives

## Converting between polar and rectangular coordinates

1 Convert the polar point $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.

2 Convert the polar equation $r = 4\cos\theta$ to rectangular form, and identify the curve.




## Identifying and sketching common polar curves

3 Identify the type of curve for $r = 2 + 2\sin\theta$, and state its key feature.

4 Identify the type of curve for $r = \cos(3\theta)$.

## The slope of a polar curve

5 State the formula for $\frac{dy}{dx}$ of a polar curve $r = f(\theta)$, using $x = r\cos\theta$ and $y = r\sin\theta$.

6 Find $\frac{dy}{dx}$ for $r = 2\theta$ at $\theta = \frac{\pi}{2}$.

## Finding horizontal and vertical tangents

7 For the circle $r = 4\cos\theta$, find the value(s) of $\theta$ in $[0, \pi]$ where the tangent line is horizontal (where $y = r\sin\theta$ has a critical point).

## Symmetry of polar curves

**8** Explain how to test whether a polar curve $r = f(\theta)$ is symmetric about the polar axis (the $x$-axis).

**9** Test whether $r = 3\sin\theta$ is symmetric about the $y$-axis (replace $\theta$ with $\pi - \theta$).

## More polar-to-rectangular conversions

**10** Convert the rectangular point $(3, 3\sqrt{3})$ to polar coordinates ($r > 0$, $0 \le \theta < 2\pi$).

**11** Convert the rectangular equation $y = x$ to polar form.

**12** Convert the polar equation $r\sin\theta = 4$ to rectangular form, and identify the curve.

## More slope-of-a-polar-curve practice

**13** Find $\frac{dy}{dx}$ for the cardioid $r = 1 + \sin\theta$ at $\theta = 0$.

**14** Find $\frac{dy}{dx}$ for the circle $r = 5$ (constant) at any angle $\theta$, and interpret the result using the fact that a circle centered at the origin has radial symmetry.

## Tangent lines at the pole

**15** Find the values of $\theta$ where the cardioid $r = 1 - \cos\theta$ passes through the pole ($r = 0$), and explain why these give the tangent line directions at the pole.

**16** Find the values of $\theta \in [0, 2\pi)$ where the rose $r = \cos(2\theta)$ passes through the pole.

## Classifying polar curves by equation type

**17** Classify each polar equation by curve type (circle, cardioid, limaçon with inner loop, rose, or line):

**a** $r = 5$

**b** $r = 3 + 3\cos\theta$

**c** $r = 1 + 3\sin\theta$

**d** $r = 4\sin(5\theta)$

## Number of petals in rose curves

**18** State the rule for the number of petals of $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$ depending on whether $n$ is odd or even.

19 How many petals does $r = 2\sin(4\theta)$ have? How many does $r = 5\cos(7\theta)$ have?

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## Graphing polar curves point-by-point

20 Complete a short table of $(\theta, r)$ values for $r = 2 + 2\cos\theta$ at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, and use it to describe the cardioid's shape.

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## Concavity considerations for polar curves

21 Explain, in terms of $\frac{dy}{dx}$, what a 'vertical tangent' means for a polar curve, and why it corresponds to $\frac{dx}{d\theta} = 0$ (with $\frac{dy}{d\theta} \neq 0$).